

Companion LXXVII: Entropic Optimal Transport and Sinkhorn Geometry of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Entropic optimal transport provides a regularized and computationally efficient variant of classical Wasserstein geometry. The resulting Sinkhorn divergence defines a smooth, stable, and differentiable metric structure on probability measures, making it particularly well-suited for topological data analysis. Building on the Wasserstein information geometry developed in Companion LXXVI, this work introduces entropic optimal transport and Sinkhorn geometry for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, inducing characteristic changes in entropic transport costs, barycentric flows, and geometric curvature. The resulting framework provides a powerful and computationally robust tool for analyzing RDCC signatures in weak-lensing topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Classical Wasserstein distances provide a natural metric on the space of diagrams, but they are computationally expensive and non-smooth. Entropic optimal transport introduces a regularization term that smooths the transport plan, yielding the Sinkhorn divergence—a fast, stable, and differentiable alternative.

In cosmology, these properties are particularly valuable: weak-lensing topology requires robust metrics that remain stable under noise, sampling variation, and small-scale fluctuations. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation mechanism modifies the gravitational potential in a scale-dependent manner, producing characteristic signatures in entropic transport geometry.

2 Entropic Optimal Transport

Given two persistence diagrams D and E with cost matrix C , the entropic OT problem is

$$\text{OT}_\varepsilon(D, E) = \min_{\pi} \left[\sum_{i,j} \pi_{ij} C_{ij} + \varepsilon \sum_{i,j} \pi_{ij} \log \pi_{ij} \right], \quad (1)$$

where $\varepsilon > 0$ is the entropic regularization parameter.

The Sinkhorn divergence is

$$S_\varepsilon(D, E) = \text{OT}_\varepsilon(D, E) - \frac{1}{2} \text{OT}_\varepsilon(D, D) - \frac{1}{2} \text{OT}_\varepsilon(E, E). \quad (2)$$

This divergence is symmetric, positive definite, and smooth.

3 RDCC Modification of Entropic Transport

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

This modifies the birth and death coordinates of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b(\ell/\ell_{\text{rel}})^{\alpha}], \quad (4)$$

$$d_{\text{RDCC}} = d [1 - \chi_d(\ell/\ell_{\text{rel}})^{\alpha}]. \quad (5)$$

The entropic OT cost becomes

$$\text{OT}_{\varepsilon}^{\text{RDCC}}(D, E) = \text{OT}_{\varepsilon}(D, E) \left[1 - \chi_{\text{OT}} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (6)$$

4 Sinkhorn Geometry in RDCC

The RDCC Sinkhorn divergence is

$$S_{\varepsilon}^{\text{RDCC}}(D, E) = S_{\varepsilon}(D, E) \left[1 - \chi_S \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (7)$$

RDCC induces:

- reduced entropic transport cost at small scales,
- enhanced cost at large scales,
- a characteristic turnover near ℓ_{rel} ,
- modified Sinkhorn barycenters,
- altered entropic geodesics in topological space.

5 Applications to Cosmological Topology

Entropic OT enables:

- stable comparison of noisy persistence diagrams,
- fast computation of barycenters across redshift bins,
- differentiable topological metrics for machine learning,
- entropic Wasserstein flows for RDCC parameter inference,
- robust RDCC vs. Λ CDM discrimination under noise.

These applications provide a computationally efficient and geometrically rich framework for analyzing RDCC signatures in weak-lensing topology.

6 Conclusion

Entropic optimal transport and Sinkhorn geometry provide a smooth, stable, and computationally efficient framework for analyzing persistence diagrams in the Relaxation-Driven Cyclic Cosmology. The relaxation mechanism modifies entropic transport costs and geometric curvature in a scale-dependent manner, producing distinctive signatures in weak-lensing topology. The present Companion establishes the foundation for future studies of entropic geometry, Schrödinger bridges, and regularized optimal transport in cosmological topology.

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